

## Assignment No. 06

// 1. Shopping Cart using let, const, and var

```
const taxRate = 0.18;
let totalPrice = 0;
var items = ["Shoes", "Bag", "Watch"];

totalPrice = 2000;

let finalPrice = totalPrice + (totalPrice * taxRate);

console.log("Items:", items);
console.log("Final Price:", finalPrice);
```

// 2. Area of Rectangle using Regular Function and Arrow Function

```
// Regular Function
function rectangleArea(length, width) {
  return length * width;
}

console.log("Area:", rectangleArea(10, 5));

// Arrow Function
const areaArrow = (length, width) => length * width;

console.log("Area:", areaArrow(8, 4));
```

// 3. Object representing a Book

```
const book = {
  title: "JavaScript Basics",
  author: "John Doe",
```

```
    year: 2024,  
  
    displayDetails() {  
        console.log(  
            `Title: ${this.title}, Author: ${this.author}, Year: ${this.year}`  
        );  
    }  
};  
  
book.displayDetails();
```

// 4. Object Destructuring

```
const car = {  
    brand: "Toyota",  
    model: "Fortuner",  
    year: 2023  
};  
  
const { brand, model, year } = car;  
  
console.log(brand, model, year);
```

// 5. Array Destructuring

```
const numbers = [10, 20, 30, 40];  
  
const [first, second] = numbers;  
  
console.log(first, second);
```

// 6. Map Method

```
const names = ["Sam", "Rahul", "Priya"];
```

```
const lengths = names.map(name => name.length);
```

```
console.log(lengths);
```

```
// 7. Filter Method
```

```
const nums = [1, 2, 3, 4, 5, 6];
```

```
const evenNumbers = nums.filter(num => num % 2 === 0);
```

```
console.log(evenNumbers);
```

```
// 8. Reduce Method
```

```
const cart = [100, 200, 300, 400];
```

```
const total = cart.reduce((sum, item) => sum + item, 0);
```

```
console.log("Total Price:", total);
```

```
// 9. Rest Operator
```

```
function sum(...numbers) {  
  return numbers.reduce((total, num) => total + num, 0);  
}
```

```
console.log(sum(1, 2, 3, 4, 5));
```

```
// 10. Spread Operator
```

```
const fruits1 = ["Apple", "Banana"];
```

```
const fruits2 = ["Orange", "Mango"];
```

```
const allFruits = [...fruits1, ...fruits2];
```

```
console.log(allFruits);
```

```
// 11. Callback Function
```

```
function delayedExecution(callback) {  
  setTimeout(callback, 2000);  
}
```

```
delayedExecution(() => {  
  console.log("Callback executed after 2 seconds");  
});
```

```
// 12. Promise Example
```

```
const myPromise = new Promise((resolve, reject) => {  
  setTimeout(() => {  
    resolve("Promise resolved after 3 seconds");  
  }, 3000);  
});
```

```
myPromise.then(message => console.log(message));
```

```
// 13. Closure Example
```

```
function outerFunction() {  
  let message = "Hello from Closure";  
  
  return function innerFunction() {  
    console.log(message);  
  };  
}
```

```
const closureFunc = outerFunction();
```

```
closureFunc();
```

```
// 14. Async/Await with Public API
```

```
async function fetchData() {  
  try {  
    const response = await fetch(  
      "https://jsonplaceholder.typicode.com/users"  
    );  
  
    const data = await response.json();  
  
    console.log(data);  
  
  } catch (error) {  
    console.log("Error:", error);  
  }  
}
```

```
fetchData();
```

```
// 15. Filter + Map + Reduce
```

```
function processNumbers(arr) {  
  
  return arr  
    .filter(num => num % 2 === 0)  
    .map(num => num * 2)  
    .reduce((total, num) => total + num, 0);  
}  
  
console.log(processNumbers([1, 2, 3, 4, 5, 6]));
```

